Parth Patel

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WORK EXPERIENCE

Software Engineer, *Deloitte* July 2022 – Oct 2024

Client: Public Health Agency of Canada

Tech stack: Python, Flask, AWS Lambda, DynamoDB, S3, FHIR

- Delivered real-time health data access to public health teams across 5 provinces by replacing a manual-pull legacy system with Python/Flask REST APIs on AWS, improving delivery speed by 3x.
- Secured 100K+ patient records and unblocked downstream data pipelines by automating PII redaction with a Python/AWS Lambda microservice, eliminating manual intervention before ingestion.
- Shipped features 4x faster and caught regressions before production by building Docker-based CI/CD pipelines on AWS with automated monitoring, cutting release time from 2 hours to 30 minutes.

Client: Employment and Social Development Canada

Tech stack: Java, Spring Boot, Python, Azure Cosmos DB

- Saved $20K/year in infrastructure costs by profiling Java/Spring Boot APIs to eliminate redundant database calls, cutting response time from 25ms to 8ms across 5,000+ daily requests.
- Reduced support resolution time by 50% for a government benefits platform by deploying a RAG system over 1,200+ tax and pension documents that powered a user-facing chatbot and replaced manual search.
- Shipped 20+ features reliably across 5 client environments by standardizing code review, staging validation gates, and rollback procedures, sustaining consistent bi-weekly delivery.

RESEARCH/PROJECTS

Research Assistant, *SJSU Research Foundation* *May 2025 - Present*

Tech stack: Python, PyTorch, Docker, MySQL, GitHub Actions

- Accelerated code feedback for 300+ students from 2-3 days to real-time by fine-tuning Qwen2.5-coder model analysis system that replaced manual TA review and scaled instructor capacity.

TensorScript – C++ Compiler for ML Computation Graphs Link

Tech stack: C++, LLVM IR, x86 Assembly

- Achieved 8x faster tensor graph execution over a NumPy baseline by building a compiler that fuses ML operators and generates cache-optimized native code, eliminating interpreter overhead.

Scout CLI – Semantic Code Search Link

Tech stack: Rust, tree-sitter, BM25, vector embeddings

- Delivered semantic cross-repo code search in under 10ms for 10,000+ functions by building Scout, an open-source CLI that fuses BM25 and vector embeddings, shipped as cross-platform binaries across 6 releases for Windows, Linux, and MacOS.

Amber – Domestic Violence Safety Platform Link

Tech stack: React Native, TypeScript, FastAPI, Python, LancChain, LangGraph, Supabase, Twilio, Groq

- Built an iOS app that detects missed check-ins and escalates emergencies by triggering Eleven Labs voice agents via Twilio voice calls to victim contacts via a LangGraph agent.

ACHIEVEMENTS

- Earned the Silver Dot Award at Deloitte for engineering a public health data platform that delivered near-real-time insights to 5 provincial agencies across Canada, serving millions of citizens.
- Won the CalHacks 2025 hackathon against 2,100 participants and 400+ projects by building an agentic AI system on Fetch.ai that autonomously coordinated tasks across distributed agents.
- Won Hack For Humanity 2026 by building Amber, an AI-powered Domestic Violence safety app that uses a LangGraph multi-agent system to detect missed check-ins and trigger emergency voice calls.

EDUCATION

San Jose State University, *Master's in Software Engineering, GPA: 3.85* *Jan 2025 – May 2026*

University of Toronto, *Bachelor of Science in Mathematics and Statistics* *Sept 2018 – June 2022*

SKILLS

Languages: Python, Java, C++, Rust, SQL

Databases: Redis, DynamoDB, MongoDB, MySQL, Oracle

Cloud/Frameworks: AWS (Lambda, RDS, S3, EC2), Flask, Spring Boot, FastAPI, Docker, Git

Certifications: Google Cloud Professional Data Engineer, Microsoft Azure Data Engineer Associate.